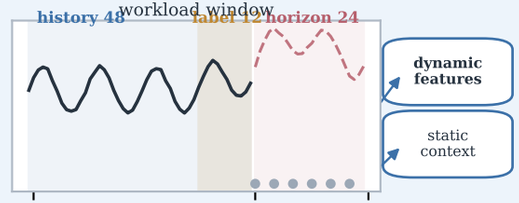
MPGFormer: Drift-Aware Multi-scale Prototype-Graph Transformer

(a) Input Workload Window



dynamic features

static context

x: [B,72,4]

future target masked

enc 48

(b) Temporal Evidence Encoder

Target RevIN → Linear + Position → Static Context Selector

mean / std control

Non-stationary Attention x2 → Short-term Conv / GRU → **time context**

history encoding: [B,48,128]

(c) Multi-scale Time-Frequency Prototype Bank

target history [B,48] → multi-scale patching → raw patches + FFT magnitudes → **learnable prototype memory** → prototype context

p=6

p=12

p=24

p=48

future padding

Prototype weights also expose uncertainty for routing.

(d) Graph-Router-Decoder Forecasting

batch-level adaptive graphs

GCN + contrastive → graph context

time context

prototype context

drift features

Drift-aware Router → w_t, w_g, w_p

weighted evidence fusion

decoder input 12 labels + 24 padding

Transformer Decoder

point forecast + quantile interval